

Simulation-Based Optimization and Cybersecurity Analysis of Warehouse Management Systems (COMP5011)

Submitted by:

KARTHIK ASHOK HONGUNTIKAR

University of Plymouth

Student ID: 10877925

karthik.honguntikar@postgrad.plymouth.ac.uk

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University of Plymouth Faculty of School of Engineering, computing, and mathematics

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Introduction:

Simul8, a powerful discrete event simulation program, is used globally for planning, designing, optimising, and re-engineering production, manufacturing, logistics, and service supply systems. It creates visual mock-ups of real processes, tests the effects of modifications, and aids decision-making.

The project aims to optimise warehouse processes using Simul8 software. The simulation model will create a virtual warehouse with resources and procedures. The goal is to assess alternative scenarios, identify bottlenecks, analyse performance, and make data-driven recommendations to enhance productivity, resource usage, and efficiency. The virtual warehouse will be created with forklift drivers at the receiving dock, and stored products collected and ready for shipping upon order placement.

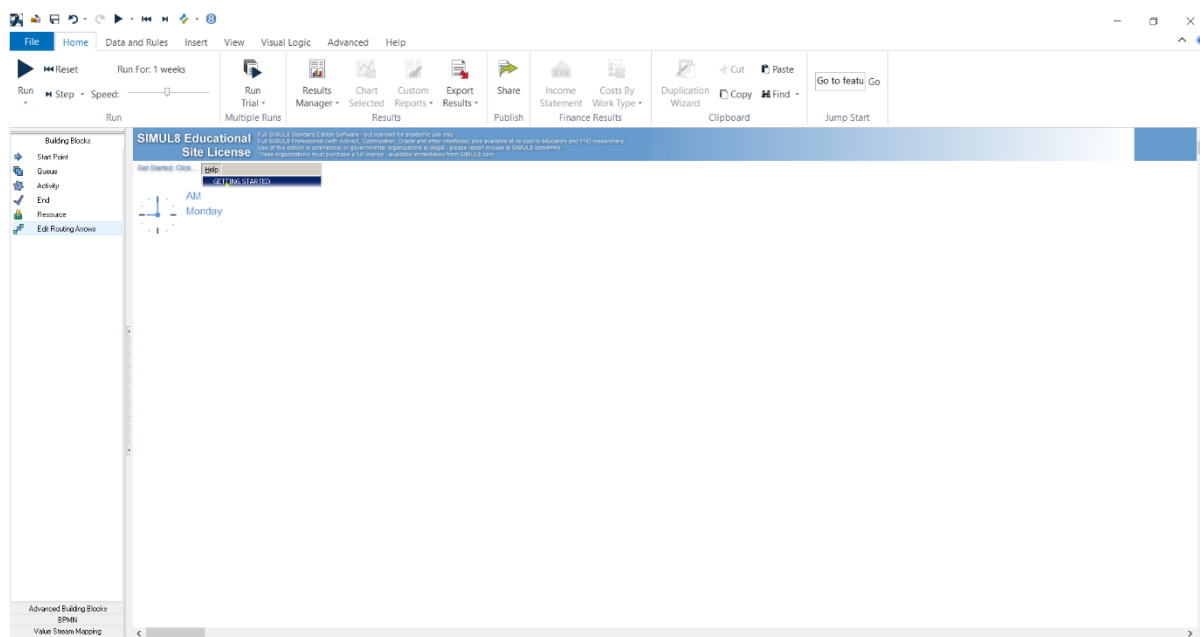


Fig 1: The simulation(simul8) window is seen in the image above.

Methodology:

The present hypothesis posits that the optimisation of warehouse operations can be achieved by the identification of bottlenecks, enhancement of resource utilisation, and implementation of effective solutions aimed at increasing system performance.

Simulation Tool: A virtual model of the warehouse activities will be created using Simul8, a potent simulation program. The development of intricate simulations made possible by Simul8 makes it possible to analyse intricate systems and procedures.

Dataset: Based on the following dataset, the simulation model is constructed:

Dimensions and layout of the warehouse

Arrival trends and inbound goods dispersion
Processing times for forklifts, storage, and unloading tasks

The quantity and qualities of the resources (forklifts, labourers, etc.) that are available
Distributions and order trends for shipments leaving the country

I am working on a comprehensive warehouse management simulation right now. It incorporates a graphic schematic that shows important locations and shows how items move from being received to being forked, stored, and shipped out. It also includes a variety of resources, like staff, storage facilities, and forklifts. It uses parameters and probability distributions that are assigned from the dataset to model order placement, processing times, and arrival patterns. Finally, it sets up data collection points to get important performance indicators, like throughput, resource utilisation, and queue lengths.

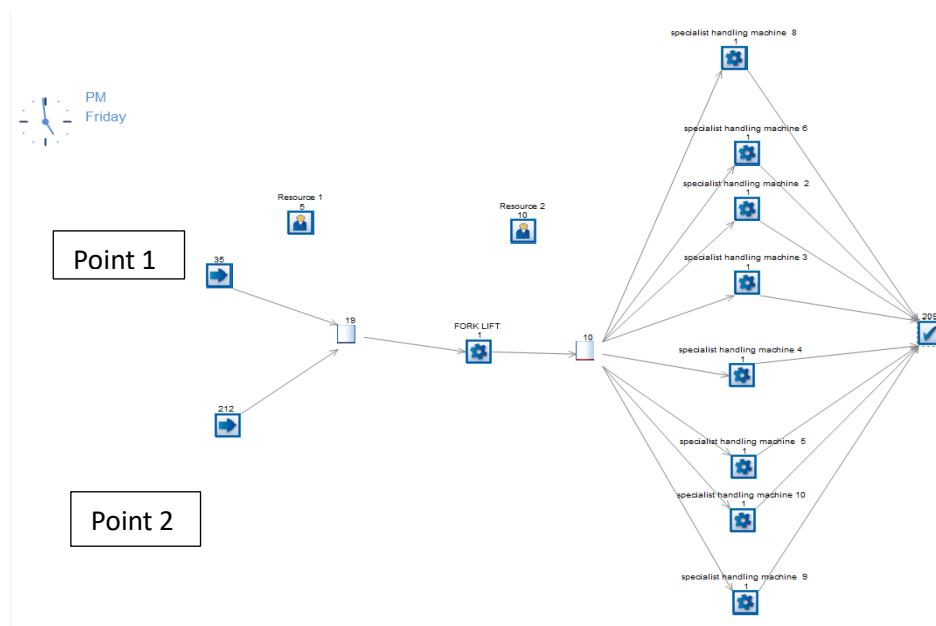


Fig 2: The above picture shows the simulation of the warehouse management system

- **Arrivals:** Goods are received from two distinct sites at the two points of arrival.
- **Resource Person 1:** This person accepts the items and makes note of all relevant information.
- **Queue 1:** Following their recording, the items are arranged in a line up beside Resource Person 1
- **Activity 1:** Items in the queue are inspected and kept close to this spot.
- **Queue 2:** Items are inspected and ready for forklifting in this second line.
- **Resource Person 2:** This person gives the forklift operators (specialist handling machine) assignments. There 8 forklift operators in the process.

The eight forklift operators are responsible for moving the cargo.

Delivery and Sorting: After the items are sorted, they are finally delivered to the agreed-upon destination.

Data analysis tools: These are used to examine the given dataset, which contains details on the warehouse's size and arrangement, arrival patterns, processing times, resource availability, and shipment distributions. Software for statistics or programming languages with libraries for data analysis could be considered tools.

The real simulation model is created using simulation software. It permits the setting up of data collecting stations, the inclusion of different resources, and the distribution of probability distributions and parameters.

Using graphic design software, a schematic of the warehouse is created that highlights key locations such as the receiving dock, storage area, and departing dock.

Communication tools: These are used to allocate and coordinate duties among warehouse employees. For example, one person can use them to assign jobs to forklift operators and to record the arrival of items.

Results:

The simulation results demonstrate how well the warehouse management system works by utilizing the integration of bulk container deliveries from ports and the careful documentation of arrival information by assigned staff. The warehouse's process flow, which includes receiving, forklift operations, storage, and shipping, is carefully planned to maximize efficiency and reduce bottlenecks. Important elements like the receiving dock, storage areas, and departure dock are clearly defined.

By integrating the parameters obtained from the dataset using probability distributions, the simulation faithfully replicates actual occurrences, impacting arrival patterns, order placement, and processing durations. This improves the prediction power of the model and guarantees an accurate portrayal of warehouse operations.

In addition, the simulation model effectively incorporates staff, storage areas, and forklifts—all crucial warehouse resources—to produce a comprehensive representation of the warehouse ecosystem. The simulation depicts the intricacies of resource allocation and queue management in the warehouse setting, with eight forklifts and two queues (one at the products receiving point and another at the items awaiting forklifting for other areas of the warehouse).

The outcomes highlight the effectiveness, utilisation and efficiency of the warehouse management system and offer insightful information about operational dynamics and optimization prospects. The warehouse may further improve throughput, reduce wait times, and maximize resource utilization through strategic resource allocation and process refinement, which will ultimately lead to operational excellence and customer satisfaction.

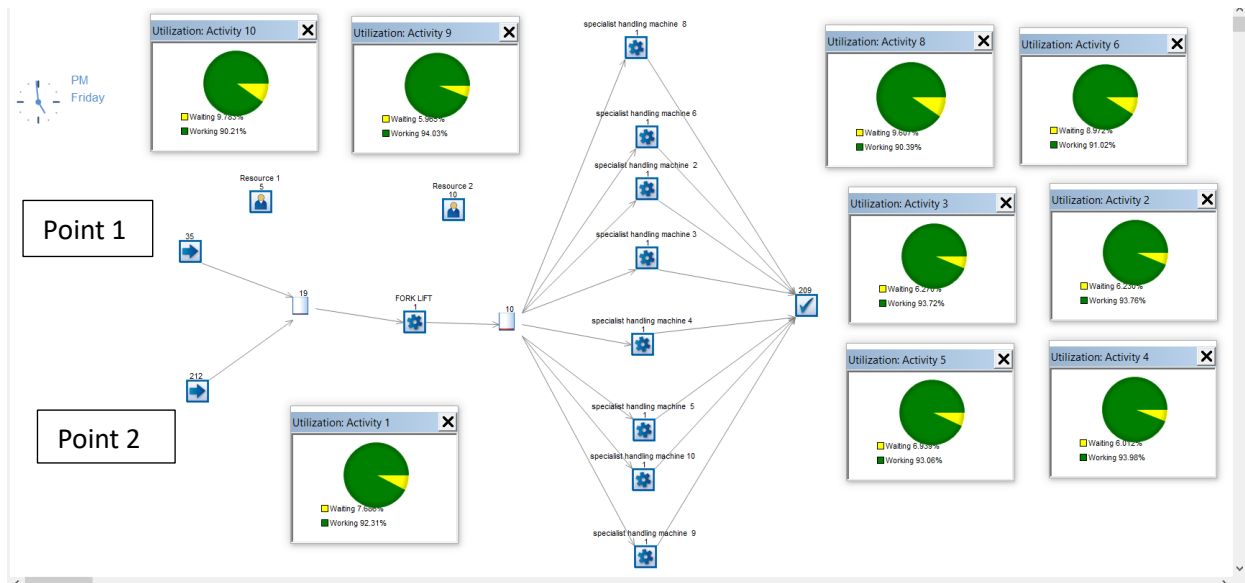


Fig3: The warehouse management system simulation with all activities operating efficiently is seen in the above image.

We obtained useful information about the effectiveness and performance of important warehouse procedures from the warehouse management simulation. Key findings from the analysis of the simulation data are as follows:

A total of 35 items entered from **Point 1**, and 212 items arrived at **Point 2**, meaning that **Queue 1** experienced an average wait time of **74.26 minutes**.

Queue 1's maximum wait time of **181.81 minutes** indicates possible areas for optimisation to lessen queue congestion.

247 items in all entered the warehouse; 19 items are currently waiting to be processed in **Queue 1**.

The forklifts' **90.1%** efficiency shows that they are using resources in material handling operations effectively.

The wait time at **Queue 2** is assessed to be **92.64** minutes on average, with a maximum wait time of **231** minutes recorded.

Queue 2 received 227 items in total, and 10 items are presently waiting to be processed.

Based on calculations, the **specialist handling machine** (forklift operator) average efficiency is **92.66%**, which suggests that the warehouse is highly productive and utilised.

Start Results

Start Point 1

☒ OK

Number of work items entered this entry point:

35

Number of work items lost:

0

Number of work items entered remainder of simulation:

35

Fig 4: The outcomes of the point 1 entry are displayed in the screenshot above.

Start Results

Start Point 2

☒ OK

Number of work items entered this entry point:

212

Number of work items lost:

0

Number of work items entered remainder of simulation:

212

Fig 5: The outcomes of the point 2 entry are displayed in the screenshot above.

Queue Results

Queue for FORK LIFT

☒ OK

Number of work items in this storage:

Currently: 19

Minimum: 0.00

Average: 7.76

Maximum: 20.00

Total Entered: 247

Graph

☒ Sync with other results
☐ Plot every change

Queuing Time:

	All	Non-Zeros
Minimum:	0.00	0.34
Average:	74.26	79.87
Maximum:	181.81	
Standard Deviation:	55.29	
Number of non zero queuing times:		212

Queuing Time within limit:

Time limit: 10 minutes

Percentage within limit: 18%

Fig 6: The outcomes of queue 1 are displayed in the screenshot above.

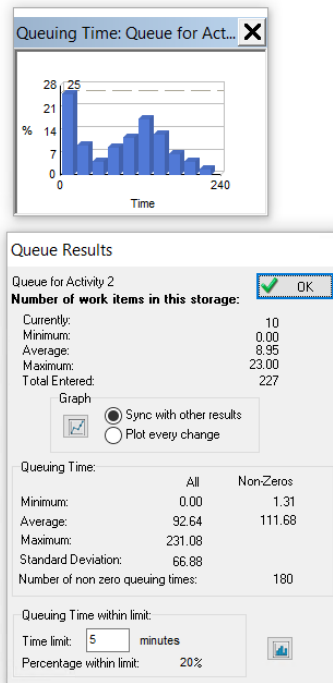
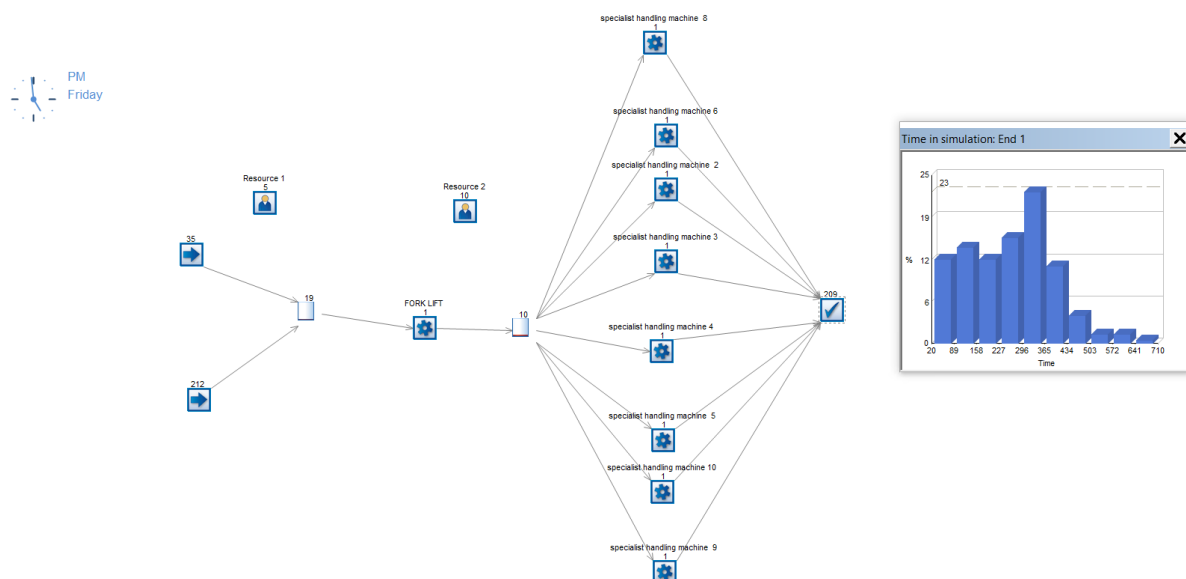


Fig 7: The outcomes of queue 1 are displayed in the screenshot above.



The simulation results show that the system has an excellent mean efficiency of 92.62%. It was found that the first and second queues had average waiting periods of 74.26 and 92.64 minutes, respectively. A few objects were seen close to both queues, but they did not interfere with the process flow or workflow. It was clear that the operation was completed without any instances of work bottlenecks. In particular, the constant flow of work combined with best practices led to an output that was higher than 90%.

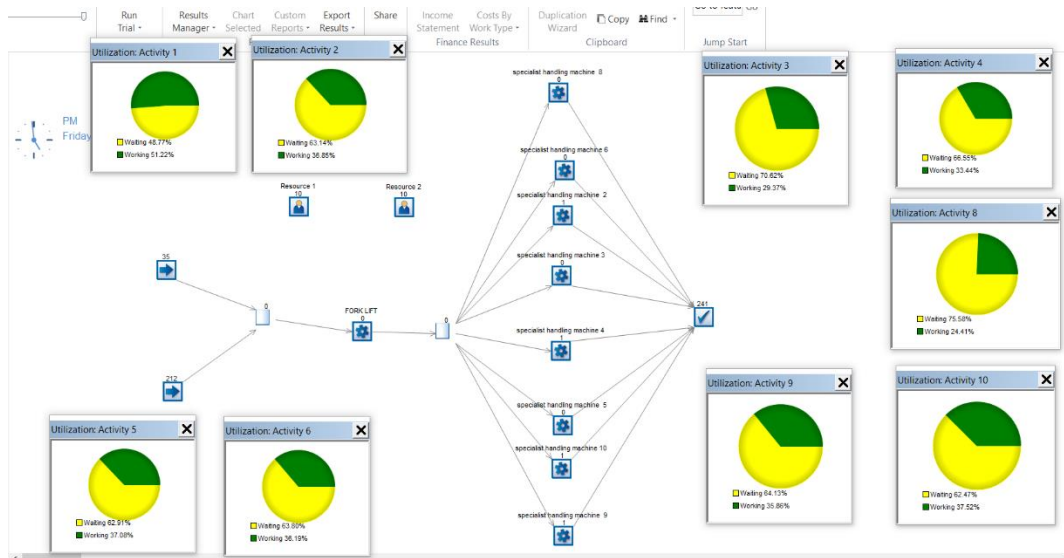


Fig 8: The simulation of a cyberattack on a warehouse management system with low activity efficiency is depicted in the above image.

The simulation results show that the warehouse management system (WMS) has sustained severe damage because of the **spear phishing attacks**. Below is a summary of the effects that were noticed:

Reduced Task Efficiency: In the wake of the cyberattack, there has been a noticeable decline in the efficiency of certain jobs inside the WMS. The simulation demonstrates this decrease in efficiency since tasks take longer to complete and resource usage is not at its best. For instance, there are negative effects on the processing times for tasks involving forklifts, storage, and unloading, which results in overall inefficiencies in warehouse operations.

Zero Queue: There was probably a backlog of tasks in the WMS that needed to be processed before the cyberattack. The queue is now zero, as the simulation results show. This indicates that there may be a backlog of work that needs to be done because the cyberattack interfered with the way tasks were completed within the system.

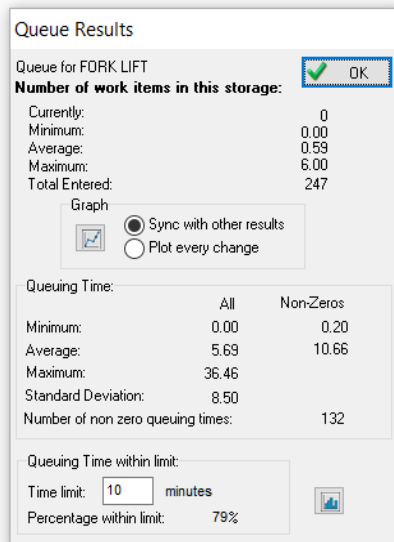
Financial Losses: The organization has experienced substantial financial losses due to a cyberattack, including direct expenses like cybersecurity equipment and forensic investigators, as well as indirect costs like missed delivery dates and inaccurate inventory.

Process Delays Outside the WMS: The cyberattack disrupts warehouse operations, causing delays in order processing, shipping, and fulfilment, potentially affecting customer satisfaction, and potentially leading to additional financial losses.

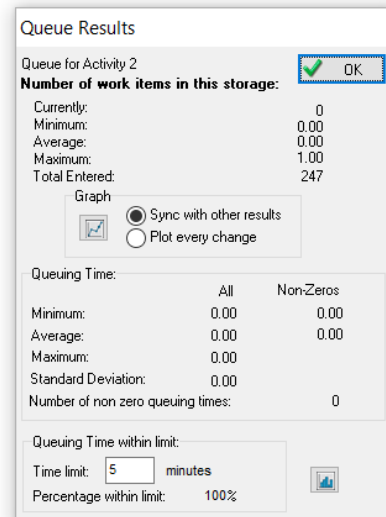
The warehouse management system (WMS) and its operations were significantly impacted by the cyberattack, as shown by the following figures:

Queue Reduction: Following the attack, both Queue 1 and Queue 2, which were previously in charge of jobs and orders, were reduced to zero.

This denotes a total stop to the processing of orders or new tasks, as well as possible interruptions to current operations.



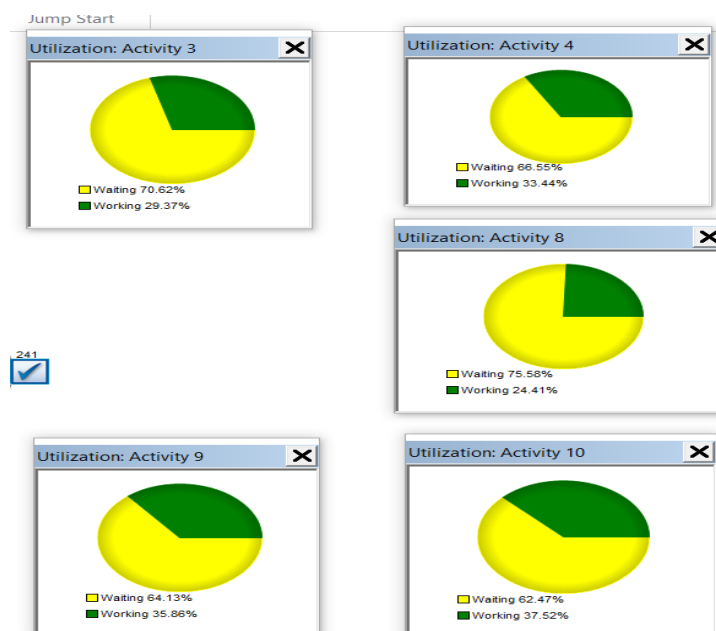
Queue 1



Queue 2

Efficiency Decline: Following the cyberattack, the forklift operators' once-high level of 92.66% efficiency has fallen to just 34.09%.

This sharp decline in productivity emphasizes how seriously it affects the workforce's capacity to carry out its responsibilities.



Efficiency after attack

Utilization and Efficiency Drop: The WMS's total 90.6% utilization and efficiency were excellent before the attack.

After the cyberattack, this percentage, however, alarmingly fell to just 35.93%.

This marked decline in efficiency and utilization points to a serious underutilization of the system and its resources, which could result in resource waste and financial losses.

To sum up, the spear phishing attacks on the warehouse management system have had a variety of negative consequences, such as decreased productivity, hiccups in operations, monetary losses, and delays in subsequent procedures. It emphasizes how crucial it is to put strong cybersecurity safeguards and incident response procedures in place to prevent these kinds of attacks and lessen their negative effects on company operations and financial results.

Discussion:

The warehouse management simulation highlights the importance of using simulation tools like Simul8 to optimize warehouse productivity. By identifying bottlenecks and optimizing resource utilization, the simulation provides insights for increased efficiency and reduced wait times. The simulation's accuracy is ensured by its close replication of real-world warehouse activities. The findings emphasize the importance of process improvement and smart resource allocation for operational excellence and customer satisfaction. Future optimization efforts should focus on increasing throughput and reducing queue congestion. The simulation aids decision-makers in developing warehouse management procedures.

A cyberattack on a warehouse management system might cause sensitive data to be stolen, operations to come to a complete stop, and data integrity to be jeopardized. The system might stop working, which would interfere with order processing, inventory control, and general warehouse operations. This would cause losses in terms of money and reputation.

Possible cyberattacks may arise from minor weaknesses in the system, such as

- i. **Phishing Attacks**
- ii. **Ransomware**
- iii. **Denial of service attack**
- iv. **Man-in-the-middle attack**
- v. **Insider threats**
- vi. **Malware**
- vii. **SQL injection**
- viii. **Social Engineering attacks**

Table 1: The relationship between security and safety systems in cybersecurity inside warehouse management systems is shown in the following simple table:

Security system	Safety system	Relationship
Access control	Physical access control	Cybersecurity access controls, including user authentication and permission, prevent illegal access to warehouse management systems. Physical access controls like locks, obstacles, and surveillance cameras protect against online and offline threats, ensuring security.
Data encryption	Emergency response plans	Data encryption protects sensitive information, while emergency response plans address safety situations like natural disasters and security breaches, reducing the negative impact on warehouse operations.
Network security	Fire safety systems	The warehouse management system is protected by network security measures like firewalls and intrusion detection, while fire safety systems like sprinklers and alarms ensure overall safety.

Table 2: The system network protocols utilized by warehouse management systems are displayed in the table below.

Protocol	Explanation
Scada	Oversees and regulates industrial machinery and procedures.
MQTT	IoT devices: a lightweight messaging protocol.
Ethernet/IP	Real-time control protocol for industrial networking.
RFID	Uses radio waves to locate and track items.

Conclusion:

In summary, this thorough analysis of cybersecurity and simulation-based warehouse management system optimization offers important new perspectives on how to protect operations and maximize efficiency. The simulation model, created with Simul8, sheds light on the dynamics of warehouse operations and identifies places where process flow and resource usage could be improved. The examination of cybersecurity flaws also emphasizes the significance of taking preventative action to lessen the dangers brought on by out-of-date software and incorrect configurations. Warehouse management systems can operate at peak efficiency while guaranteeing the security and safety of vital processes by combining simulation tools, data analysis, and cybersecurity rules. Warehouse management operational excellence and customer happiness will be enhanced by ongoing efforts to improve security protocols, streamline workflows, and respond to new threats.

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